

simplex spanned by the endmembers (fully-constrained least squares, FCLS; implemented as δ -augmented NNLS followed by renormalisation), so the coefficients are proper abundance fractions over the convex hull. The dominant endmember of every pixel emits $Q - 1$ tokens, so the document length grows with Q .

$$g_b(x_d) = \arg \max_{k \in \{0, \dots, Q-1\}} \pi_k \mathcal{N}(x_{db} \mid \mu_k, \sigma_k^2), \quad (9)$$

$$n_d^{\text{V12}}[(b, g)] = \mathbb{1}[g_b(x_d) = g], \quad |V_{\text{V12}}| = BQ,$$

where $(\pi_k, \mu_k, \sigma_k^2)_{k < Q}$ is a one-dimensional Gaussian mixture fitted once per scene to the unnormalised band intensities of the sampled pixels (a random subset of 50 000 values), with the components indexed in increasing order of μ_k . Each band emits one token, as in V3, but the bins are shared by all pixels and follow the intensity distribution instead of splitting each spectrum's own range into equal intervals.

$$w_b = \left\lfloor \frac{\text{MI}(x_b; y)}{\max_{b'} \text{MI}(x_{b'}; y)} Q_{\max} \right\rfloor, \quad (10)$$

$$n_d^{\text{V20}}[(b, q)] = w_b \mathbb{1}[q_b(x_d) = q].$$

The token alphabet is V3's, $|V_{\text{V20}}| = BQ$ (=1600 at $B = 200, Q = 8$; mean ≈ 1376 across the six labelled scenes since B varies). Bands with $w_b = 0$ emit nothing, so the *effective* (non-empty-column) vocabulary is smaller still (≈ 770 mean).

$$\text{F-1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2 P_c R_c}{P_c + R_c}. \quad (11)$$

$$\text{NPMI}(w_i, w_j) = \frac{\log \frac{p(w_i, w_j) + \epsilon}{p(w_i)p(w_j)}}{-\log(p(w_i, w_j) + \epsilon)}, \quad (12)$$

$$c_v = \frac{1}{N} \sum_i \cos(\mathbf{v}_i, \sum_j \mathbf{v}_j),$$

with $\mathbf{v}_i = (\text{NPMI}(w_i, w_\ell))_{\ell=1}^N$ over the top- N topic words.

$$\hat{z}_d = \arg \max_k \theta_{dk},$$

$$\text{F-7} = \frac{I(\hat{Z}; Y)}{H(Y)} = U(Y \mid \hat{Z}) = 1 - \frac{H(Y \mid \hat{Z})}{H(Y)}. \quad (13)$$

$$\text{F-14} = \frac{2}{K(K-1)} \sum_{k < k'} \frac{|T_k \cap T_{k'}|}{|T_k \cup T_{k'}|}, \quad T_k = \text{top-}N(\phi_k). \quad (14)$$

$$\text{F-18} = \frac{1}{\binom{S}{2}} \sum_{s < s'} \frac{1}{K} \max_{\sigma \in \mathfrak{S}_K} \sum_k \cos(\mathbf{u}_k^{(s)}, \mathbf{u}_{\sigma(k)}^{(s')}), \quad (15)$$

over S reseeded refits, with $\mathbf{u}_k^{(s)}$ the top- N indicator vector of topic k in run s .

$$\text{F-22}(d) = \min_{\delta \in \mathbb{Z}^{|V|}} \|\delta\|_1 \text{ s.t.}$$

$$\arg \max_k \theta_k(n_d + \delta) \neq \arg \max_k \theta_k(n_d), \quad (16)$$

$$n_d + \delta \geq 0,$$

reported as the per-scene median, capped at a sentinel $\text{MAX_STEPS} = 50$ when no flip is found.

$$\text{aligned}(d) = \mathbb{1} \left[|\text{top}_N(n_d) \cap \text{top}_N(\phi_{z_d})| \geq \tau \right. \\ \left. \vee \text{top}_3(n_d) \cap \text{top}_N(\phi_{z_d}) \neq \emptyset \right], \quad (17)$$

$$\text{decisive}(d) = \mathbb{1} \left[\text{aligned}(d) = 1 \vee |\text{supp}(n_d)| \geq 3 \right],$$

$$\text{F-15} = \sum_d \text{aligned}(d) / \sum_d \text{decisive}(d).$$

$$p_{dv} = \frac{n_{dv}}{\sum_{v'} n_{dv'}},$$

$$N_{\text{eff}}(d) = \exp(H(p_d)) = \exp\left(-\sum_v p_{dv} \log p_{dv}\right). \quad (18)$$

N_{eff} counts the tokens that effectively carry the mass of p_d (or of a topic's ϕ_k): it is low when the mass is concentrated on few tokens and at most $|V|$.

$$\mathcal{A}(V_i) = \frac{1}{|\mathcal{B}|} \sum_{m \in \mathcal{B}} \frac{c_v^{(m)}(V_i) - \min_j c_v^{(m)}(V_j)}{\max_j c_v^{(m)}(V_j) - \min_j c_v^{(m)}(V_j)} \in [0, 1], \quad (19)$$

averaged over backbones $\mathcal{B} = \{\text{LDA}, \text{HDP}, \text{ProdLDA}, \text{ETM}\}$; $\mathcal{A} \rightarrow 1$ when a recipe sits near the per-backbone top everywhere.

I. INTRODUCTION

The companion papers P1 [1] and P3 [2] hold the topic model backbone fixed (online-VB LDA) while varying the *wordification recipe* that converts a pixel spectrum into a multiset of LDA-consumable tokens. P3 shows that the recipe matters on every axis it reports from the twelve-axis evaluation framework introduced in P1, with V12 (Gaussian-mixture-component tokens) reaching the highest mean F-2 and F-7 under LDA while winning two of the six scenes on each.

This paper inverts the question. We hold the recipe *set* $\{\text{V1}, \dots, \text{V20}\}$ constant and vary the backbone across four topic-model families: LDA (baseline carried over from P3), Hierarchical Dirichlet Process (HDP) [3], Product LDA (ProdLDA) [4], and the Embedded Topic Model (ETM) [5]. All four share the basic generative structure (topics are distributions over a discrete vocabulary, documents are mixtures of topics). LDA and HDP place a Dirichlet and a Dirichlet-process prior on the topic proportions. The model run under the ProdLDA label is the mixture-decoder form of Srivastava and Sutton's amortised topic model (their NVLDA), with a standard normal prior on the topic logits; the product-of-experts decoder that defines ProdLDA was not run (§II). ETM shares that prior and the amortised encoder and parameterises the topic-word distribution as a softmax of embedding dot products, so the two neural backbones differ only in the decoder.

A. The headline finding

The wordification ranking is *not* invariant under the backbone choice. Table I summarises the per-backbone F-2 coherence winner across the six labelled-scene panel. LDA and

TABLE I
PER-BACKBONE F-2 c_v WINNER ON THE SIX LABELLED-SCENE PANEL
(UNIFORM / $Q = 8$, MEAN ACROSS SCENES).

Backbone	Best recipe	Mean c_v
LDA	V12 (GMM-token)	0.88
HDP	V7 (absorption triplet)	0.615
ProdLDA	V3 (joint band-bin)	0.863
ETM	V12 (GMM-token)	0.816

ETM agree on V12; HDP picks V7; ProdLDA picks V3. Three of four backbones disagree on the single best recipe.

This is the formal version of the observation that the wordification choice and the backbone choice cannot be made independently. P3’s recommendation that V1 retain canonical-default status assumed an LDA backbone; this paper shows that switching backbone changes the practical canonical too.

B. Contributions

- 1) A 4×19 factorial sweep over four topic-model backbones (LDA, HDP, ProdLDA, ETM) and nineteen wordification recipes (V1–V20; V16 scaffolded only) on six labelled HSI scenes. 456 F-2 cells, 456 F-14 cells.
- 2) Three empirical backbone–recipe affinities on F-2: LDA and ETM pick V12, with V3 and V20 close behind, HDP picks the sparse-event recipe V7, and ProdLDA picks V3. ETM and ProdLDA share their prior and encoder, so their disagreement comes from the decoder parameterisation or from optimisation, which this study does not separate.
- 3) A per-recipe *affinity score* (19) that measures how a recipe transfers across the four backbones, identifying V12 (top of the coherence affinity ranking, V20 second) and V13 (lowest affinity, at or near the per-backbone minimum) as the extremes.
- 4) A revised practical recommendation: the (backbone, wordification) pair is a single hyper-decision and should be reported jointly.

II. THE FOUR BACKBONES

All four backbones are formally defined by the canonical equation set shared across the paper series: LDA by (1), HDP by (2), ProdLDA by (3), and ETM by (4). The subsections below give only the hyperparameters and implementation that specialise each generic backbone to this study.

A. Online-VB LDA (P1, P3)

The canonical fixed-K LDA (1) with online variational Bayes inference [6]. Symmetric Dirichlet priors $\alpha = 0.45$ on topic mixtures and $\eta = 0.20$ on topic-word distributions. K chosen by the same policy as P3, from the mean document length $\bar{L}$ of each recipe and scene:

$$K = \begin{cases} K_{P1}, & \bar{L} \geq 8, \\ \text{clip}(\text{round}(\bar{L}/2), 3, K_{P1}), & 2.5 \leq \bar{L} < 8, \\ 4, & \bar{L} < 2.5, \end{cases}$$

where $K_{P1} = \max(4, \min(12, n_{\text{classes}}))$. Implementation: `sklearn.LatentDirichletAllocation`, random seed 42.

B. Hierarchical Dirichlet Process (HDP)

Bayesian nonparametric topic model with two-level stick breaking (2) [3]. We use `gensim’s HdpModel` with truncation $T = 20$ at the corpus level and $K = 8$ at the document level. Random seed 42. The number of effective topics is data-driven; the topic *rank* (the strength of each topic’s prior weight) is part of the inference output.

C. Product LDA (ProdLDA)

The model run under the ProdLDA label is the amortised variational topic model of Srivastava and Sutton [4] in its mixture-decoder form (their NVLDA), (3): a standard normal prior $\delta_d \sim \mathcal{N}(0, I)$ on $\delta_d \in \mathbb{R}^K$, $\theta_d = \text{softmax}(\delta_d)$, $w_{dn} \sim \text{Cat}(\sum_k \theta_{dk} \beta_k)$ with $\beta_k = \text{softmax}(\omega_k)$ from a free $K \times |V|$ matrix of logits ω_k . The product-of-experts decoder that defines ProdLDA was not run. Encoder: two hidden layers of 100 units producing the Gaussian posterior over δ_d ; Pyro implementation, Adam, 80 epochs, batch 128, learning rate 10^{-3} , random seed 42.

D. Embedded Topic Model (ETM)

Joint word–topic embedding model [5] with the same $\mathcal{N}(0, I)$ prior and the same amortised encoder as the ProdLDA-labelled model. Its decoder is $\beta_k = \text{softmax}(\rho^\top \alpha_k)$ of (4), with word-embedding matrix $\rho \in \mathbb{R}^{L \times |V|}$ ($L = 64$), per-topic embedding $\alpha_k \in \mathbb{R}^L$ and the mixture likelihood of Dieng et al. The two neural backbones therefore differ only in the decoder parameterisation: a free $K \times |V|$ topic–word matrix against the rank- L factorisation $\rho^\top \alpha_k$. With $L = 64$ and $K \leq 12$ the factorisation does not restrict which topic–word matrices can be represented, so the two decoders differ in parameterisation, not in capacity. Same 80-epoch / batch-128 / learning-rate- 10^{-3} / seed-42 protocol.

III. METHOD: FACTORIAL DESIGN

The factorial design is fully crossed: for each pair (backbone, recipe) we use *the same* pre-computed doc-term matrix as in P3 (so the wordification stage is identical across backbones) and refit the backbone from scratch. For each cell we report:

- F-2 c_v coherence (12) on the top-10 words per topic, using the same estimator as P1 (c_v : sliding-window NPMI with cosine aggregation).
- F-14 mean off-diagonal top-10 jaccard between topics (14) (lower = more diverse).

All recipes share the per-row min–max normalisation and uniform quantiser of (5); the five recipes with dedicated equations are V1 (band-frequency, (6)), V3 (joint band-bin, (7)), V8 (NFINDR endmember-fraction, (8)), V12 (GMM-token, (9)) and V20 (MI-weighted bands, (10)).

F-7 NMI is also reported under each non-LDA backbone (§IV-C); F-1 classification, F-22 counterfactual and F-18 reliability are reported for the LDA backbone only (carried over from P3), because they would require a separate downstream classifier pipeline per backbone, which is left to future work.

IV. RESULTS

A. F-2 c_v factorial

Table II reports the mean F-2 c_v across the six labelled scenes for every (backbone, recipe) pair, and Fig. 1 shows the same matrix as a heatmap.

V20 (mutual-information-weighted bands) ranks second under LDA, third under ETM, fourth under ProdLDA and, leaving out the flagged V15 cell (eleventh with it), tenth under HDP, where the leaders are V7 (absorption triplets, 0.61) and V17 (sparse-coding dictionary atoms, 0.59), followed by V1 (0.54). Under LDA / ETM V12 wins; under ProdLDA V3 wins; V20 trails the per-backbone winner by about 0.06 c_v on average (excluding HDP). The cross-backbone affinity ranking below quantifies this.

B. Per-recipe affinity score

We define a recipe’s *cross-backbone affinity score* $\mathcal{A}(V_i)$ by (19): the average, over the four backbones $\mathcal{B} = \{\text{LDA}, \text{HDP}, \text{ProdLDA}, \text{ETM}\}$, of the within-backbone min–max-normalised c_v of recipe V_i . $\mathcal{A} \in [0, 1]$; higher means the recipe consistently scores near the per-backbone top across all four backbones. Table III reports $\mathcal{A}$ computed directly from the full 19×4 cells of Table II, so every recipe, including the multi-scale V14 (continuous-wavelet tokens), the graph-Laplacian V18 and the label-aware V20, sits on one common scale (Fig. 2).

V12 has the highest cross-backbone affinity ($\mathcal{A} = 0.76$), with V20 (0.75), V1 (0.74) and V3 (0.73) close behind. On the within-backbone normalised scale, V12, V20 and V3 score between 0.80 and 1.00 under LDA, ProdLDA and ETM, and V1 between 0.64 and 0.82; under HDP the four score 0.10 (V12), 0.32 (V20), 0.76 (V1) and 0.01 (V3). V20 is the only *label-aware* recipe in this leading group, making it the most cross-backbone-portable label-aware recipe in the sweep and a near-replacement for V12 / V3 when label information is available. V3, although it only wins coherence outright under ProdLDA, remains a top-4 affinity recipe because it is consistently strong everywhere except HDP. **V13 (learned VQ-VAE) has the lowest affinity** ($\mathcal{A} = 0.01$), at or near the per-backbone minimum in all four: none of the four backbones gives the reconstruction-optimal VQ-VAE codebook coherent topics. Note that V8 scores low here (0.09) on *coherence* affinity specifically: it is a small-vocabulary recipe whose strength is label coupling (F-7), not c_v ; the F-7 factorial below ranks it very differently.

C. F-7 NMI under each backbone: V8 cross-backbone leadership

We compute F-7, the label-normalised mutual information $I(\hat{Z}; Y)/H(Y)$ (13) (the asymmetric uncertainty coefficient

$U(Y | \hat{Z})$, not the symmetric NMI), between the argmax topic per document and the ground-truth labels for the six labelled scenes, repeated under each backbone. The procedure is: (1) fit the backbone via its deterministic builder (LDA with the per-recipe K of §II; ProdLDA and ETM with $K = K_{P1}$ for every recipe in these F-7 runs; HDP with its 20-topic truncation), (2) project each sampled labelled pixel through the fitted model to obtain the topic mixture θ_d , (3) take $z_d^* = \arg \max_k \theta_{d,k}$, (4) compute the F-7 score $I(z^*; y)/H(y)$ of (13). For HDP we use gensim’s `HdpModel.__getitem__` corpus projection; for ProdLDA and ETM the trained encoder produces θ directly. The result is a $19 \times 4 = 76$ -cell matrix of six-scene means; Table IV shows the twelve recipes with the highest four-backbone mean (48 cells). The per-scene values of all 76 cells are in the code repository under `data/derived/v_sweep/f7_topic_to_label/` (LDA) and `data/derived/v_sweep/backbones_f7/` (HDP, ProdLDA, ETM).

Fig. 3 shows the per-scene F-7 winner under each backbone. The F-7 backbone factorial reveals *three distinct cross-backbone families*:

a) *Family 1: consistently strong*: V8, V20, V2. V8 is top-5 under every backbone, shares with V2 the smallest cross-backbone standard deviation ($\sigma = 0.06$ for both) and leads at mean 0.431 (winning outright only under ProdLDA). V2 holds that small spread at a lower level (7th under LDA and HDP); V20 is strong on average but less cross-backbone-consistent ($\sigma = 0.12$, 6th under HDP) and is the unique label-aware recipe here. All three have long documents and take $K = K_{P1}$ under every backbone, so their comparison is not affected by the topic-count difference discussed next.

b) *Family 2: apparent backbone specialists*: V11 (product quantisation) wins HDP (0.571) and ETM (0.530) but collapses to 0.292 under LDA and 0.088 under ProdLDA, the widest cross-backbone spread in the sweep. V19 (UMAP coordinates) shows a similar HDP-favoured profile. Both recipes emit three or four tokens per document and take $K = 3$ under LDA, which caps F-7, bounded by $\log K/H(Y)$, at a six-scene mean of 0.46; their HDP and ETM means exceed that cap (§IV-E).

c) *Family 3: LDA-dominant*: V12, V3, V14. Top-5 under LDA but middle of the pack under the other backbones. V12’s F-7 LDA win disappears entirely under HDP / ETM (where V11 wins) and gets edged out by V8 / V2 under ProdLDA.

The take-away is that the backbone factorial swap dimension matters *as much as the recipe dimension*: V11 looks like a poor choice under LDA but is the headline winner under HDP / ETM, and V12’s LDA leadership does not generalise. This corroborates the P3 finding that no recipe-backbone choice is canonical.

D. Three backbone–recipe affinities

The F-2 factorial (Table II) shows three empirical affinities. They describe which recipe each backbone favours; the runs do not identify why.

LDA and ETM: dense GMM tokens. LDA and ETM both pick V12 (GMM-token, 1600 word types from the $B \times Q$ joint

TABLE II

F-2 c_v MEAN ACROSS THE SIX LABELLED SCENES, FULL 19×4 FACTORIAL (HDP, ProdLDA AND ETM COVER V1..V15, V17..V20). THE HDP VALUE FOR V15 (0.52) WAS COMPUTED ON THE STORED V15 WORDIFICATION THAT HOLDS THE $Q = 32$ VOCABULARY (192 TOKENS; 96 ON PAVIA U); EVERY OTHER CELL USES THE $Q = 8$ VOCABULARY (V15: 48 TOKENS; 24 ON PAVIA U). THE V15 CELL OF FIG. 1 AND V15'S AFFINITY IN TABLE III INHERIT IT.

Backbone	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20	best
LDA	0.64	0.44	0.84	0.45	0.43	0.53	0.29	0.32	0.71	0.33	0.23	0.88	0.25	0.63	0.36	0.42	0.57	0.38	0.85	V12
HDP	0.54	0.44	0.31	0.44	0.44	0.32	0.61	0.31	0.35	0.35	0.50	0.34	0.31	0.31	0.52	0.59	0.42	0.50	0.41	V7
ProdLDA	0.75	0.44	0.86	0.43	0.43	0.53	0.51	0.31	0.73	0.33	0.34	0.83	0.25	0.47	0.37	0.42	0.48	0.37	0.74	V3
ETM	0.66	0.44	0.79	0.43	0.44	0.55	0.26	0.31	0.73	0.32	0.24	0.82	0.25	0.62	0.36	0.43	0.58	0.35	0.77	V12

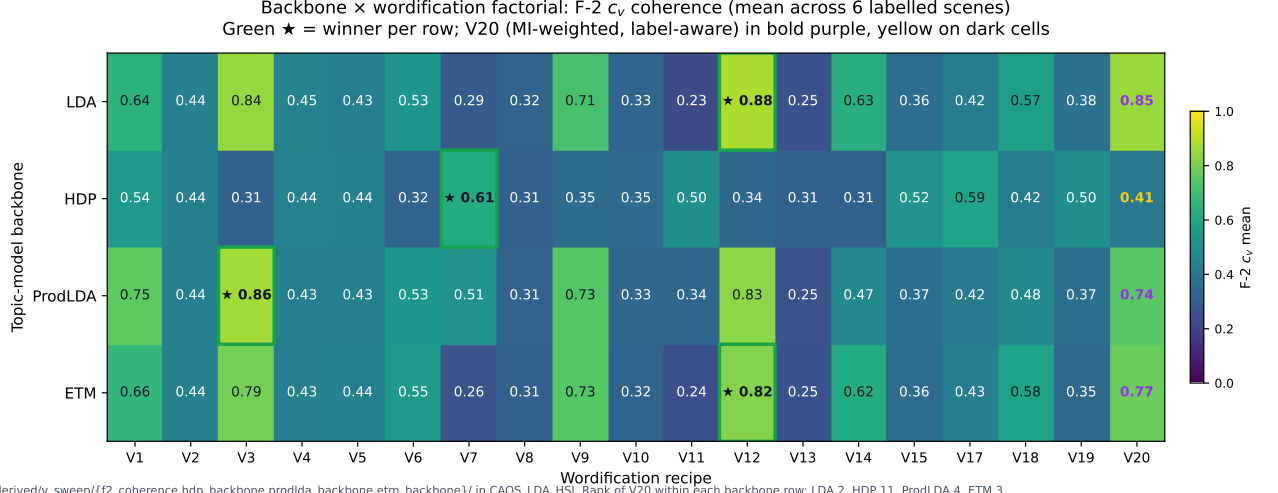


Fig. 1. F-2 c_v across the 19×4 backbone factorial: a single heatmap with one row per backbone and the recipes in index order. $\star$ and a green border mark each row's winner (bold cell in Table II); V20's value is in bold purple (yellow on dark cells), and the footer gives V20's rank within each row (the HDP rank counts the flagged V15 cell of Table II; without it V20 is tenth). The winner moves between recipes (V12 for LDA / ETM, V3 for ProdLDA, V7 for HDP), making the recipe ranking backbone-dependent.

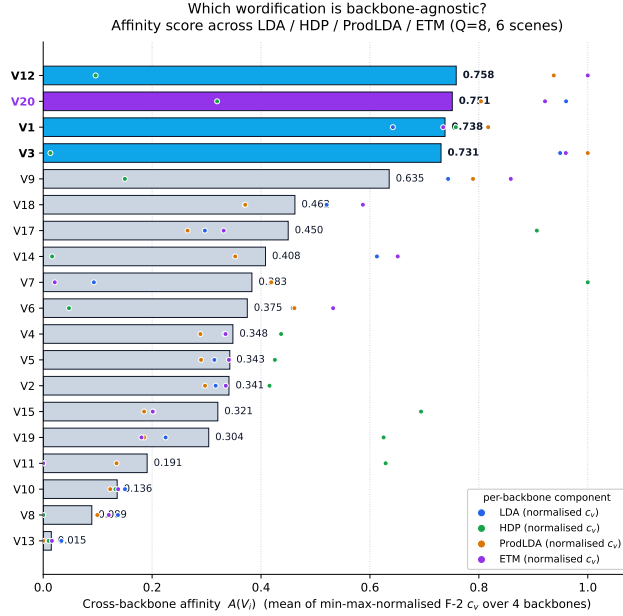


Fig. 2. Cross-backbone affinity $A(V_i)$ (19) per recipe (Table III). V12 tops the ranking (0.76), V20 second (0.75); V13 has the lowest affinity (0.01), at or near the per-backbone minimum. V20 is the only label-aware recipe in the leading group.

vocabulary with $B = 200$, $Q = 8$), with V3 and V20 close behind (LDA: V12 0.88, V20 0.85, V3 0.84; ETM: V12 0.82,

TABLE III
CROSS-BACKBONE AFFINITY SCORES $\mathcal{A}(V_i)$ (19), COMPUTED FROM THE FULL 19×4 FACTORIAL OF TABLE II (MIN-MAX NORMALISED WITHIN EACH BACKBONE, THEN AVERAGED).

Recipe	$\mathcal{A}$
V12	0.76
V20	0.75
V1	0.74
V3	0.73
V9	0.64
V18	0.46
V17	0.45
V14	0.41
V7	0.38
V6	0.37
V4	0.35
V2	0.34
V5	0.34
V15	0.32
V19	0.30
V11	0.19
V10	0.14
V8	0.09
V13	0.01

TABLE IV
F-7 NMI MEAN ACROSS THE SIX LABELLED SCENES FOR THE TOP-12 RECIPES RANKED BY 4-BACKBONE MEAN. BOLD = PER-BACKBONE WINNER.

Recipe	LDA	HDP	ProdLDA	ETM	Mean
V8 (NFINDR)	0.463	0.451	0.328	0.482	0.431
V20 (MI-weighted)	0.520	0.356	0.221	0.490	0.397
V2 (intensity-bin)	0.453	0.347	0.324	0.456	0.395
V11 (product quantisation)	0.292	0.571	0.088	0.530	0.370
V12 (GMM-token)	0.534	0.220	0.238	0.488	0.370
V3 (joint band-bin)	0.524	0.200	0.169	0.443	0.334
V19 (UMAP coord)	0.286	0.530	0.051	0.394	0.315
V15 (spectral indices)	0.313	0.438	0.058	0.449	0.315
V13 (VQ-VAE)	0.311	0.399	0.113	0.406	0.307
V14 (CWT-Morlet)	0.457	0.220	0.114	0.430	0.306
V18 (graph-Laplacian)	0.428	0.198	0.080	0.330	0.259
V1 (band-frequency)	0.455	0.081	0.091	0.381	0.252

V3 0.79, V20 0.77).

HDP: sparse-event tokens. HDP picks V7 (absorption triplets, at most six tokens per document; 0.61), followed by V17 (0.59).

ProdLDA: the joint (band, bin) vocabulary. ProdLDA picks V3 (joint (band, bin) vocabulary, $B \times Q = 1600$ types; 0.86), with V12 second (0.83).

ETM and the ProdLDA-labelled model share the $\mathcal{N}(0, I)$ prior and the amortised encoder (§II); the only implemented difference is the decoder parameterisation. Their different winners (V12 against V3) therefore come from the decoder parameterisation or from optimisation, and no ablation in this study separates the two.

E. V11: winner under HDP and ETM, weak under LDA and ProdLDA

V11 (product quantisation, $M = 4$ sub-vectors $\times K_s = 8$ codewords) has a vocabulary of $MK_s = 32$ tokens, fifty times smaller than V12's $BQ \approx 1600$; only V2 ($Q = 8$), V8 (its

6–12 endmembers), V10 and V19 ($3Q = 24$) have smaller vocabularies. A small vocabulary fares very differently under each backbone, but the backbones also differ in topic count, and F-7 cannot exceed $\log K/H(Y)$. Under LDA, V11's four-token documents take $K = 3$ (§II), which caps F-7 at 0.40 to 0.61 across the six scenes (six-scene mean 0.46); V11 reaches 0.77 to 0.82 of that cap on Botswana and Salinas. The F-7 runs of ETM use $K = K_{P1}$ (12 on four scenes) and HDP keeps up to 20 topics, so V11's 0.571 under HDP and 0.530 under ETM are out of reach of any three-topic fit. The contrast below therefore mixes the backbone with the topic count:

- **Under HDP**, V11 is the best recipe (0.571). HDP keeps all of its 20 truncation topics for V11 (effective number 19.9 to 20.0 in the F-2 runs), against 1.2 to 6.2 for V12. Across the nineteen recipes HDP's F-7 follows the number of topics it keeps (Spearman $\rho = 0.59$), and the recipes on which it keeps one or two topics (V1, V4, V5, V6) score F-7 near zero.
- **Under ETM** (embedded decoder), the topic-word distribution is $\beta_k = \text{softmax}(\rho^\top \alpha_k)$ of (4), with word embeddings $\rho \in \mathbb{R}^{L \times |V|}$ and topic embedding $\alpha_k \in \mathbb{R}^L$; for V11, ρ is $L \times 32$. V11 is the best recipe under ETM (0.530).
- **Under LDA**, V11 reaches only 0.292, below the 0.46 cap of its three topics; within that cap, topic-word distributions that overlap strongly over only 32 tokens may further blur the argmax topic, which the runs do not measure.
- **Under ProdLDA**, V11 collapses to 0.088, near chance, although the ProdLDA-labelled model shares ETM's prior and encoder. The two differ only in the decoder parameterisation, so the collapse comes from the free topic-word matrix or from optimisation; the runs do not separate the two.

V19 (UMAP coords, vocab = 24) follows the same pattern: it is second under HDP at 0.530, after V11 (0.571), but collapses to 0.051 under ProdLDA. The recipe-backbone choice is two-dimensional.

V. PRACTICAL RECOMMENDATION

The pair (backbone, wordification) is a *single* hyperdecision in LDA-on-HSI work, not two independent choices. Reporting V12 as the best recipe is incomplete without specifying that the result holds for LDA and ETM; reporting V3 as the best recipe is incomplete without specifying ProdLDA.

We make three recommendations:

- 1) Future LDA-on-HSI papers should report the (backbone, wordification) pair explicitly in table headers, not as separate hyperparameters.
- 2) Cross-backbone affinity (19) is a more informative summary than per-backbone-winner counts: it identifies recipes that generalise across families (V12, V20, V1, V3) versus those that only win in narrow regimes (V7 for HDP, V13 nowhere).
- 3) V1 (band-frequency) remains a defensible *reproducibility* canonical because of P3's F-18 result, but should not be claimed as a universal best on coherence in any backbone.

Per-scene F-7 NMI winner under each backbone: 6 labelled scenes × 4 backbones

Colour = recipe family of the wordification taxonomy



Fig. 3. Per-scene F-7 NMI winner under each backbone. Tile colour encodes the winning recipe’s family from the wordification taxonomy. HDP’s top winner is V11 (4/6 scenes; V13 on Indian Pines, V19 (manifold family) on Salinas); ETM also picks V11 on 3/6 scenes. LDA has the most diverse per-scene winners. ProdLDA favours V8 / V12 / V2.

VI. LIMITATIONS

- LDVAE-T [7], a spectral-unmixing VAE with a Dirichlet-prior latent (an abundance/endmember model, not an off-the-shelf topic-model classification backbone), was considered as a candidate fifth backbone but has no public implementation; adapting it to the wordification factorial and adding its row is left to future work.
- The backbone labelled ProdLDA is the mixture-decoder form of (3); the product-of-experts ProdLDA was not run, and no ablation separates the decoder parameterisation from optimisation in the ETM–ProdLDA comparison.
- F-1 (Bayesian classification) is only run under LDA. Running F-1 under HDP / ProdLDA / ETM requires extending the topic-routed classifier pipeline to each backbone, which is left to future work.
- HDP’s $K_{\text{effective}}$ is extracted as the number of topics holding at least 1% of corpus mass under the top-level stick-breaking weights (the `hdp_to_lda` expected topic prevalence), feeding F-16 model-selection adequacy $|K_{\text{effective}} - \# \text{classes}|$; F-16 is shown in the interactive web application rather than tabulated here.

DATA AVAILABILITY

Per-cell results (one JSON record per scene, recipe and backbone) are archived in the code repository

https://github.com/fsantibanezleal/CAOS_LDA_HSI under `data/derived/v_sweep/: f2_coherence/ and f7_topic_to_label/` for LDA, `hdp_backbone/`, `prodllda_backbone/` and `etm_backbone/` for F-2 and F-14 under the other three backbones, and `backbones_f7/` for their F-7.

REFERENCES

- [1] F. Santibáñez-Leal, “Beyond accuracy: A multi-axis evaluation framework for interpretable topic models on hyperspectral imagery,” Preprint, Zenodo, 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.21504115>
- [2] F. Santibáñez-Leal, “Which wordification matters? A nineteen-recipe sweep of the interpretable-topic-model framework on hyperspectral imagery,” Preprint, Zenodo, 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.21504117>
- [3] Y. W. Teh, M. I. Jordan, M. J. Beal, and D. M. Blei, “Hierarchical Dirichlet processes,” *Journal of the American Statistical Association*, vol. 101, no. 476, pp. 1566–1581, 2006.
- [4] A. Srivastava and C. Sutton, “Autoencoding variational inference for topic models,” in *International Conference on Learning Representations (ICLR)*, 2017, arXiv:1703.01488. [Online]. Available: <https://openreview.net/forum?id=BybtVK9lg>
- [5] A. B. Dieng, F. J. R. Ruiz, and D. M. Blei, “Topic modeling in embedding spaces,” *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 439–453, 2020.
- [6] M. D. Hoffman, D. M. Blei, and F. R. Bach, “Online learning for latent Dirichlet allocation,” in *Advances in Neural Information Processing Systems 23 (NIPS 2010)*, 2010, pp. 856–864. [Online]. Available: <https://papers.nips.cc/paper/3902-online-learning-for-latent-dirichlet-allocation>

- [7] G. Giannetti and F. Z. Qureshi, "Latent Dirichlet transformer VAE for hyperspectral unmixing with bundled endmembers," *arXiv preprint arXiv:2511.17757*, 2025. [Online]. Available: <https://arxiv.org/abs/2511.17757>